Mini-project SMA: Stochastic Processes for Environmental Data

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Main concepts and goals of this mini-project

Final objective:

- Predict time series in the future
- Create an ensemble of artificial forecasts

AI-driven approach:

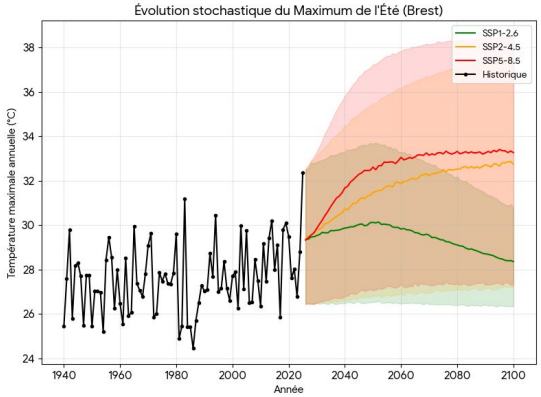
- Code it using PyTorch and GPU
- Let AI help you

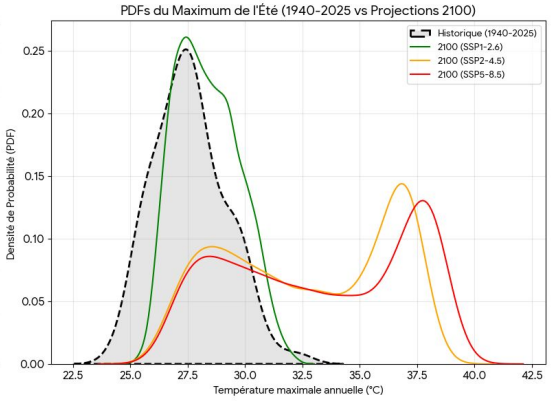
Stochastic modeling:

- Use recent neural networks (e.g., diffusion models)
- Condition it to an external variable (CO2)

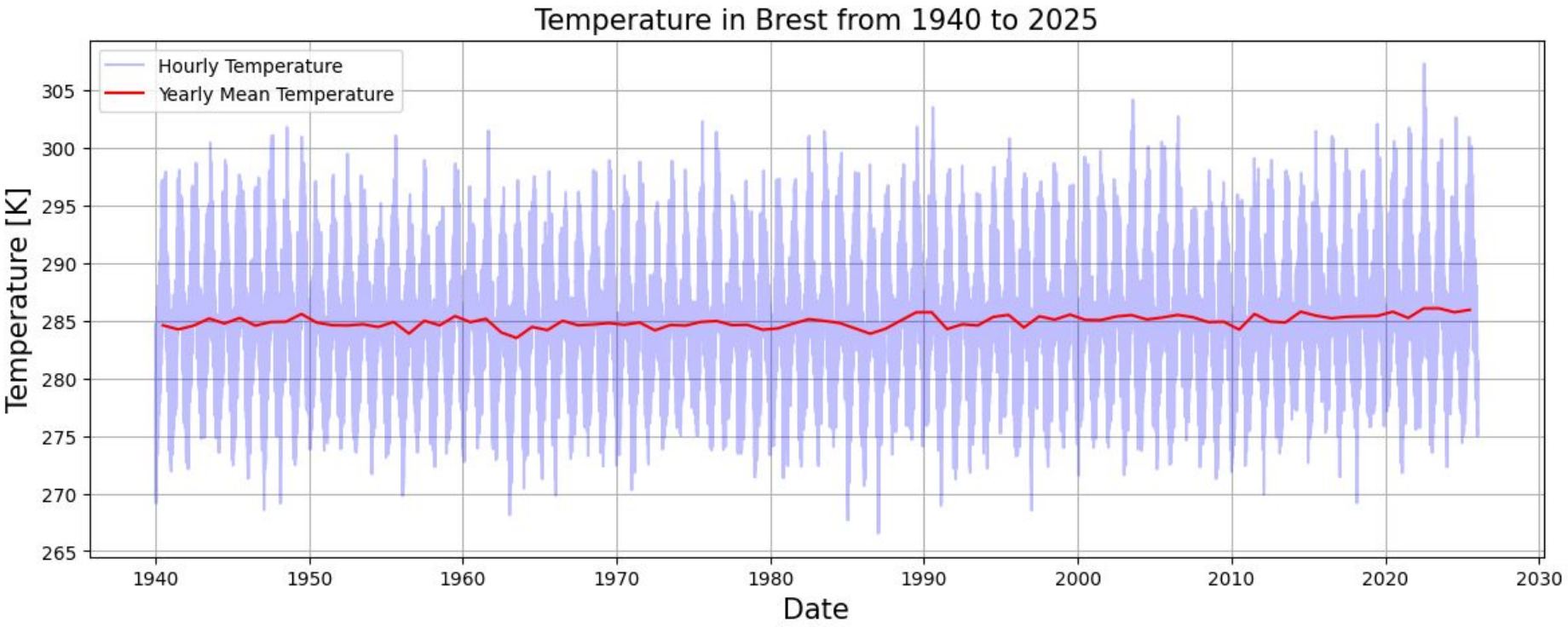
Climate application:

- Real environmental data (temperatures)
- Compute statistics of extremes in the future

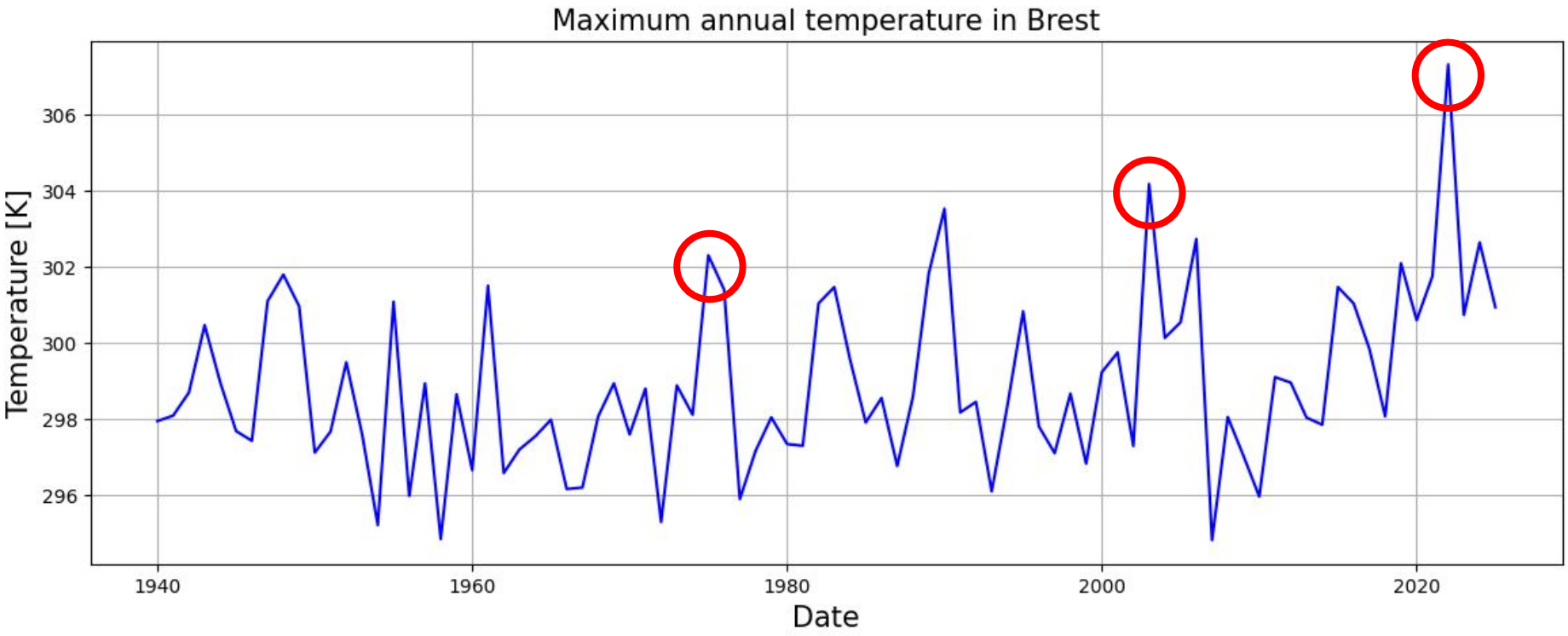




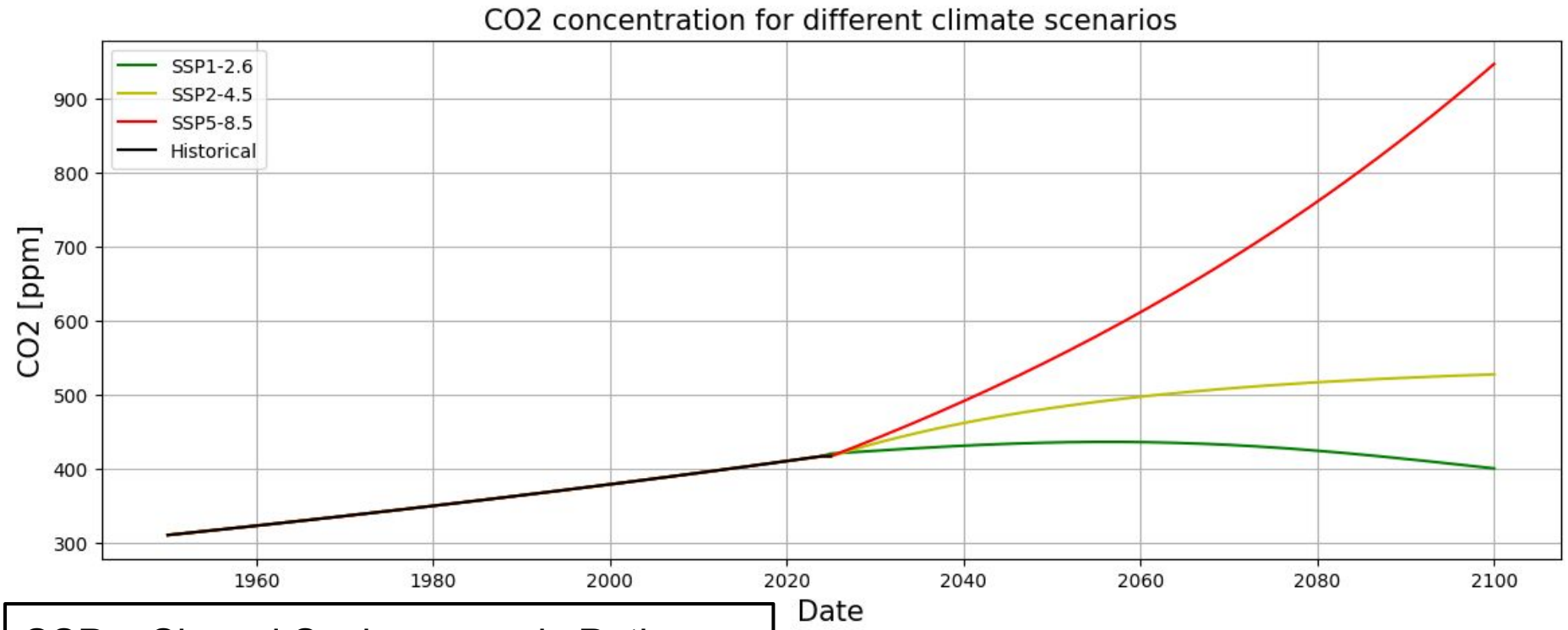
Data: air temperatures in Brest from 1940 to 2025



Focus on the annual maximum of air temperatures

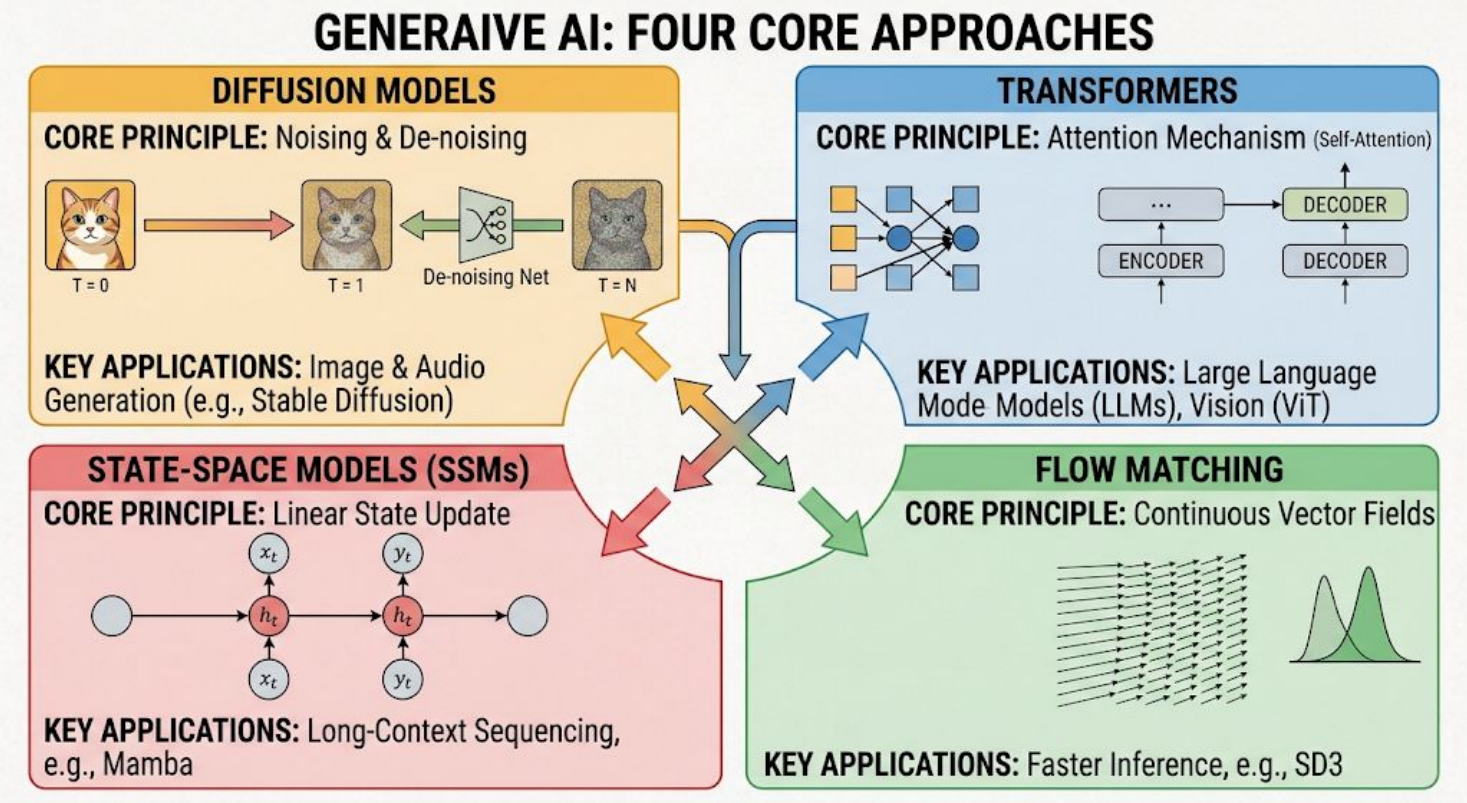


Future CO2 emissions and climate scenarios



SSP = Shared Socioeconomic Pathways

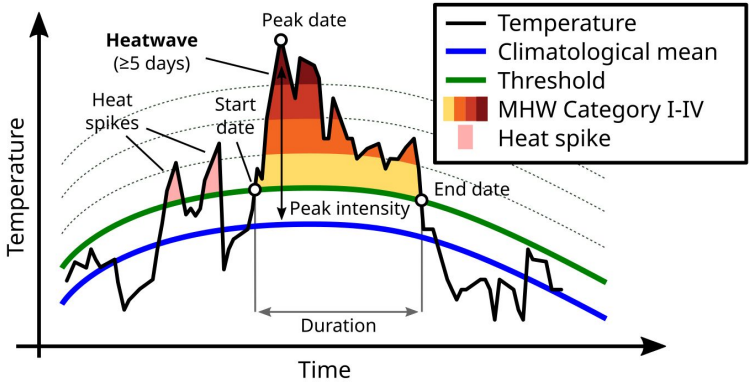
Recent machine learning methods for generative AI



A Jupyter Notebook with 2 parts

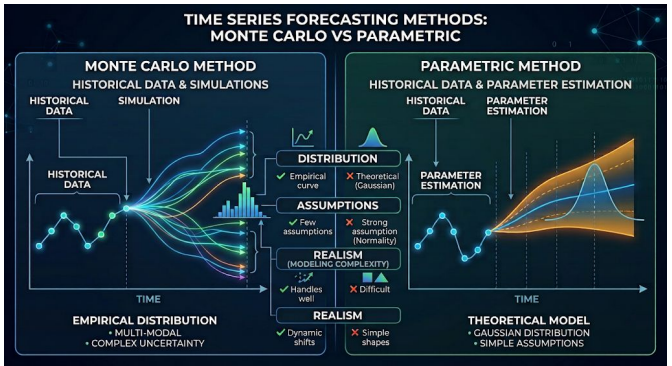
Part 1: Statistical analysis

- Seasonal cycle:
  - distribution of the daily temperature
  - over a typical year
  - extract the 90% quantile
- Intensity-duration plot:
  - identify extreme events (heat waves)
  - exceeding the 90% quantile
  - during several consecutive days



Part 2: Machine learning

- Probabilistic forecasts:
  - make probabilistic predictions in the future
  - use Monte-Carlo or parametric methods
  - condition your forecasts on CO2 emissions
- Question and additional study:
  - what is the probability to exceed 40°C in Brest, at the horizon 2100?
  - make the same analysis with the minimum annual temperatures



Evaluation of this mini-project

- **Quality of the code:**
  - must be concise
  - comments are welcome
- **Quality of the outputs:**
  - you are dealing with physical variables
  - do not forget the units (e.g., °C, ppm)
- **Explanation of the method:**
  - you must describe mathematically the chosen approach
  - write it directly into the notebook using Latex
- **Discuss your results:**
  - results have physical meaning
  - you should be able to comment the outputs

Let’s start

- Download the Jupyter **Notebook** here
- Download the three **datasets** here
- Use Google Colab to share the notebook with your **five colleagues**
- Upload **one notebook per group** to Moodle here
- Deadline is Friday, **October 30**, 2026